Fungal NGS Pipeline

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Year: 2026

Introduction

Next-generation sequencing (NGS) workflows transform raw sequencing reads into alignment and quality-control outputs that can be inspected and reused in downstream analyses. This project presents a modular Nextflow workflow for processing sequencing data associated with Candida auris, alongside Python utilities for summarising alignment quality, genome coverage and estimated sequencing depth. The project combines workflow automation, reference-genome handling, command-line bioinformatics tools, tabular summaries and interpretation of overrepresented sequences. Its primary purpose is to provide a reproducible processing workflow and a transparent summary of technical outputs.

Project Objective and Analytical Questions

This is primarily a computational workflow and quality-control project rather than a definitive biological study. It addresses how sequencing data for a defined set of Candida auris samples can be processed reproducibly; how well reads map to a selected reference assembly; what coverage and estimated-depth summaries show; and how sequence-level inspection and BLAST results can support interpretation of overrepresented sequences.

Dataset and Reference Genome

The six-sample cohort consists of SRA accessions SRR10059687, SRR10059688, SRR10059689, SRR10059690, SRR10059691 and SRR10059692. A separate accession, SRR26185932, was used for a technical workflow task and is not treated as a seventh sample in the six-sample cohort. The reference is the Candida auris B8441 V2 assembly, accession GCA_002759435.2. The local reference directory contains the assembly sequence and annotation resources, plus derived indexes/databases used by relevant workflow steps. Large raw sequencing and derived index files are kept locally rather than included in the public GitHub repository. Raw datasets can be retrieved by accession from the NCBI Sequence Read Archive (SRA) or the European Nucleotide Archive (ENA). The repository documents accessions and download sources rather than bundling raw sequencing files.

Methods and Workflow

The repository contains three Nextflow workflow definitions: workflow/main.nf (main workflow), workflow/resistant_cohort.nf (resistant cohort workflow) and workflow/bedtools_coverage.nf (standalone BEDTools coverage workflow). Python utilities in scripts/ produce summaries of alignment quality, coverage and depth: alignment_qc.py, coverage_summary.py and depth_summary.py. The project also includes sample metadata (metadata/samples.tsv), automated tests (test/test_pipeline.py), a Nextflow configuration file and software-version information in docs/tool_versions.txt.

Software and Reproducibility

The workflow is designed for a Linux/WSL2 environment and uses Nextflow to coordinate execution. Relevant workflow steps use established command-line bioinformatics tools including HISAT2, SAMtools and BEDTools. BLAST resources support sequence-level investigation of selected overrepresented sequences. Reproducibility is supported by workflow definitions, configuration, sample metadata, dependency documentation, tests and recorded tool versions. Raw SRA/FASTQ files, reference indexes, BLAST database files, Nextflow cache, execution logs and backup files are local working resources and should not be uploaded as part of the public source repository. Validation during project review included successful Nextflow preview checks for the three workflow entry points and a Python unit-test run in which all six tests passed. A preview checks workflow construction; it does not execute the full analysis.

Results

1. Alignment summary

The six cohort samples showed mapping rates between 85.78% and 91.78%, with a mean of approximately 89.92%. Mapping rate describes the proportion of reads mapped to the selected reference; by itself, it does not establish biological quality or confirm the origin of every mapped read.

Sample	Mapping rate
SRR10059687	91.78%
SRR10059688	89.47%
SRR10059689	85.78%
SRR10059690	91.41%
SRR10059691	89.75%
SRR10059692	91.30%
Mean	**89.92%**

2. Coverage summary

The coverage summary contains 5,583 features per sample. Mean breadth of coverage is approximately 99.87%–99.91%, with three to five genes per sample having zero covered bases. These values should be interpreted in the context of the reference annotation and the summary method.

3. Depth summary

Estimated mean depth is approximately 384×–746×. This estimate was calculated from mapped bases divided by reference genome length; it is not a direct per-position depth calculation and should be described as an approximate average.

4. Overrepresented sequences

The results directory contains selected overrepresented sequences in FASTA format, BLAST output tables, an exact-counts table and an interpretation document. These outputs support review of the sequence-level investigation; biological claims should be limited to what the sequences and database matches support.

Discussion

The project demonstrates a practical workflow for organizing NGS processing and producing compact, reviewable quality-control summaries. Mapping rates are broadly consistent across the six cohort samples, while coverage and estimated-depth summaries provide complementary views of the mapped

data. Overrepresented-sequence analysis adds a diagnostic layer for sequences that may warrant further inspection. Several limitations should remain explicit. A successful Nextflow preview validates workflow construction but does not prove that every process completes in a full run. The reported depth is an estimate based on mapped bases and genome length, not a per-base depth distribution. High mapping and breadth-of-coverage values are technical metrics and should not by themselves be interpreted as evidence for a specific biological phenotype or resistance mechanism.

Conclusions

This project combines Nextflow workflow automation with Python-based summaries for a six-sample Candida auris cohort. Outputs include alignment-QC, coverage and estimated-depth tables, plus files supporting investigation of overrepresented sequences. Workflow definitions, tests, metadata and tool-version records provide a foundation for reproducible execution and future extension. The project demonstrates practical skills in NGS workflow orchestration, Linux/WSL2, Nextflow, Python scripting, reference-data management, quality-control interpretation and reproducible project organization.

Repository Structure

The public repository should contain source code, configuration, metadata, tests, documentation and compact summary results. Large raw sequencing files, reference indexes, BLAST databases, Nextflow cache/logs and temporary or backup files should remain local.

```
01_Fungal_NGS_Pipeline/
├── workflow/ (three .nf files)
├── scripts/ (three Python scripts)
├── test/test_pipeline.py
├── metadata/samples.tsv
├── data/raw/README.md
├── data/reference/README.md
├── results/summary/ (summary tables and sequence-inspection outputs)
├── docs/tool_versions.txt
├── nextflow.config
├── requirements.txt
├── .gitignore
└── README.md
```

Data Sources and Documentation

1. NCBI Sequence Read Archive: https://www.ncbi.nlm.nih.gov/sra
2. European Nucleotide Archive: https://www.ebi.ac.uk/ena/browser/home
3. NCBI Assembly / Datasets, GCA_002759435.2: https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_002759435.2/
4. Nextflow documentation: https://www.nextflow.io/docs/latest/
5. HISAT2 documentation: https://daehwankimlab.github.io/hisat2/
6. SAMtools documentation: https://www.htslib.org/doc/samtools.html
7. BEDTools documentation: https://bedtools.readthedocs.io/

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